

Self-Entangled Duality Dynamics (SED):

From Spacetime Separation to Dual Emergence

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Abstract

This paper establishes Self-Entangled Duality Dynamics (SED) as the logical successor to Self-Entangled Geometric Dynamics (SEGD). SEDG's failure stems from its presupposition of independent spatial Ψ_s and temporal Ψ_t degrees of freedom, thereby smuggling a classical spacetime background into a theory whose central aim is to eliminate precisely such backgrounds. SED resolves this ontological paradox through the dual spacetime state construction $\Psi_{\text{SED}} = \Psi_{st} \otimes \Psi_{ts}$.

This framework yields a novel intrinsic origin for spectral asymmetry: unlike Fano interference arising from discrete-continuum coupling, the SED asymmetry parameter ϵ derives from the q -deformed commutator norm $\mathcal{C} = \|[H_{st}, H_{ts}]_q\|$ of spacetime operators with their duals. We derive the SED-corrected spectral line shape, the integral definition of ϵ , and its scaling relations; predict that quantum critical point (QCP) systems will spontaneously exhibit Exceptional Points (EPs); and demonstrate that the Hamiltonian's odd-even decomposition leads to critical exponent splitting $\nu_{\text{odd}} \neq \nu_{\text{even}}$ near QCPs, yielding three falsifiable experimental benchmarks distinguishing SED from Fano effects and standard non-Hermitian optics.

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1 Introduction: From Spacetime Separation to Dual Emergence

1.1 The Structural Defect of SEG D

Self-Entangled Geometric Dynamics (SEG D) attempted to extend quantum entanglement to spacetime itself, positing the core ansatz:

$$\Psi_{\text{SEG D}} = \Psi_s \otimes \Psi_t \quad (1)$$

where $\Psi_s \in \mathcal{H}_s$ and $\Psi_t \in \mathcal{H}_t$ belong to purportedly independent spatial and temporal Hilbert spaces.

The failure of SEG D is *structural*, not technical. Equation (1) conceals a fatal logical circularity: the tensor product form presupposes that space and time exist as separable subsystems *prior to any dynamics*. This is equivalent to smuggling a classical fixed spacetime stage into a theory whose raison d'être is the elimination of precisely such stages. SEG D describes fundamentally inseparable spacetime quantum gravity using a “pre-separated spacetime basis”—a paradigm that contradicts itself at the ontological level. No perturbative correction or parameter optimization can rescue this.

1.2 The SED Paradigm Shift: Dual Spacetime as Primitive

SED abandons the independent spacetime subsystem assumption entirely, establishing an irreducible dual spacetime tensor structure as the theory's first axiom.

Axiom 1.1 (SED Primitive State).

$$\boxed{\Psi_{\text{SED}} = \Psi_{st} \otimes \Psi_{ts}} \quad (2)$$

The physical content of the symbols:

- Ψ_{st} : the spacetime-integrated *explicit-order state*, describing the spacetime-coupled system from the conventional observational perspective;
- Ψ_{ts} : the spacetime-flipped *dual latent state*, representing the dual object when the observational relation is inverted;
- The exchange $st \leftrightarrow ts$ is not a simple label permutation but a complete inversion of the subject-object duality relation. Tensor order carries irreducible physical directionality.

1.3 Three Core Paradigm Features of SED

1. **Background Independence:** No pre-given spacetime geometric background exists; spacetime itself emerges dynamically from the st/ts duality relation.
2. **Intrinsic Causal Directionality:** $H_{st} \otimes H_{ts} \neq H_{ts} \otimes H_{st}$ generates an intrinsic arrow of time without additional thermodynamic assumptions.
3. **Universal Falsifiability:** Duality flips produce an intrinsic spectral asymmetry parameter ϵ , amplified by coherent volume near quantum critical points to experimentally observable regimes.

2 The Internal Clock: Emergence of Time from Duality Flips

Before presenting the algebraic formalism, we must address the most primitive question: *what counts as “time” in a theory with no external time parameter?*

2.1 The Two-Beat Cycle

In SED, time is not a background parameter $t \in \mathbb{R}$. It is an *emergent count* generated by the $st \leftrightarrow ts$ flip itself. Define the internal clock τ as:

$$\tau \in \mathbb{Z}_2 = \{0, 1\} \quad (3)$$

where $\tau = 0$ corresponds to the st -dominant phase and $\tau = 1$ to the ts -dominant phase. The dynamics is a discrete two-beat process:

$$\hat{H}_{\text{tick}}(\tau) = \begin{cases} H_{st} \otimes H_{ts} & \text{if } \tau \equiv 0 \pmod{2} \\ H_{ts} \otimes H_{st} & \text{if } \tau \equiv 1 \pmod{2} \end{cases} \quad (4)$$

Critical observation: τ is not an external time t . It is a *purely combinatorial count* of flips—dimensionless, scaleless, merely “which beat.” In the thermodynamic limit, the discreteness of τ is smoothed over, but its mod-2 structure survives as a $\mathbb{Z}_2$ -grading.

2.2 The Arrow of Time from Flip Asymmetry

Define the *duality asymmetry entropy*:

$$\Delta S_{\text{arrow}} \sim \ln \left| \frac{\langle \Psi | H_{st} \otimes H_{ts} | \Psi \rangle}{\langle \Psi | H_{ts} \otimes H_{st} | \Psi \rangle} \right| \quad (5)$$

This quantity is:

- **Zero** for reversible processes: the two beats are perfectly symmetric, $st \rightarrow ts$ and $ts \rightarrow st$ occur with equal probability.
- **Positive** for irreversible processes: the flip probabilities are unbalanced, generating a directed flow.

Equation (5) is not a metaphor. It is SED’s fluctuation theorem. The parameter $\mathcal{C}$ (to be defined in Section 3) serves as a novel quantum resource; its dissipation directly generates the thermodynamic arrow of time.

2.3 Why Standard Tensor Products Fail

In standard quantum mechanics, $\mathcal{H}_A \otimes \mathcal{H}_B \cong \mathcal{H}_B \otimes \mathcal{H}_A$ via the swap operator P , which is unitary and an *inner automorphism*. This means $A \otimes B$ and $B \otimes A$ are physically indistinguishable.

In SED, $H_{st} \otimes H_{ts}$ and $H_{ts} \otimes H_{st}$ have *mutually dependent basis definitions*:

- $|i\rangle_{st}$ is defined only with $|j\rangle_{ts}$ as background;

- $|j\rangle_{ts}$ is defined only with $|i\rangle_{st}$ as background.

This circular definition destroys the standard swap operator P . P is not an inner automorphism but an *outer automorphism*—or more precisely, it does not exist as a well-defined operator in the algebra. This is the physical origin of the III_1 factor structure to be developed in Section 3.2.

3 Mathematical Framework: The Type III_1 Factor and q -Deformed Cyclic Algebra

3.1 The SED Dual Hamiltonian and Unique Odd-Even Decomposition

The fundamental duality Hamiltonian governing SED dynamics is:

$$\hat{H}_{\text{SED}} = \hat{H}_{st} \otimes \hat{H}_{ts} \quad (6)$$

Splitting the tensor into symmetric and antisymmetric parts under order exchange yields the *unique* orthogonal odd-even decomposition:

$$H_{\text{even}} = \frac{1}{2} (H_{st} \otimes H_{ts} + H_{ts} \otimes H_{st}) \quad (\text{Hermitian, governs spectral center, static energy}) \quad (7)$$

$$H_{\text{odd}} = \frac{1}{2} (H_{st} \otimes H_{ts} - H_{ts} \otimes H_{st}) \quad (\text{anti-Hermitian, governs directionality, spectral asymmetry}) \quad (8)$$

The total physical Hamiltonian:

$$H_{\text{phys}} = H_{\text{even}} + H_{\text{odd}} \quad (9)$$

Key distinction: H_{even} only shifts the central frequency; H_{odd} does not shift the center but distorts the line profile, introducing asymmetry.

3.2 Why the Algebra is Type III_1 : The Tomita-Takesaki Theorem

The mutual definability of $|i\rangle_{st}$ and $|j\rangle_{ts}$ implies that the observables of the st -sector and the ts -sector are not statistically independent. In algebraic quantum field theory terms, the von Neumann algebra $\mathcal{M}$ generated by SED observables satisfies:

$$\mathcal{M} \cong \mathcal{M} \otimes \mathcal{B}(\mathcal{H}) \quad \text{and} \quad \mathcal{M}' \cong \mathcal{M}' \otimes \mathcal{B}(\mathcal{H}) \quad (10)$$

where $\mathcal{M}'$ is the commutant. This is the defining property of a **Type III factor**.

More precisely, SED's algebra is a **Type III_1 factor**, characterized by:

1. **No trace:** There exists no faithful normal semi-finite trace on $\mathcal{M}$. The standard quantum mechanical trace $\text{Tr}(\cdot)$ fails because st and ts are not independent subsystems whose probabilities can be multiplied.

2. **Modular automorphism group:** By the Tomita-Takesaki theorem, every faithful normal state ω on $\mathcal{M}$ defines a one-parameter automorphism group σ_t^ω :

$$\sigma_t^\omega(A) = \Delta^{it} A \Delta^{-it}, \quad A \in \mathcal{M} \quad (11)$$

where Δ is the modular operator. In SED, this modular flow *is* the physical time evolution. The parameter t in σ_t^ω is not external time but the *thermal time* generated by the state itself.

3. **Connes' classification:** For Type III₁, the spectrum of Δ is $\mathbb{R}_+$. This means there is no discrete “ground state”—the duality flips generate a continuous flow, consistent with the smoothed two-beat cycle in the thermodynamic limit.

Physical interpretation: In standard quantum mechanics (Type I), we assume an external classical observer who can perform measurements on independent subsystems. In SED (Type III₁), the observer is *inside* the algebra—there is no “outside” from which to trace over one subsystem. The $st \leftrightarrow ts$ flip is not a unitary operation we perform; it is the *intrinsic modular flow* of the state itself.

3.3 The q -Deformed Cyclic Duality Algebra

The standard quantum tensor product satisfies exchange equivalence $\mathcal{H}_A \otimes \mathcal{H}_B \cong \mathcal{H}_B \otimes \mathcal{H}_A$ with unitary swap. In SED, as established above, the standard swap fails. The algebraic structure is instead a *cyclic algebra* with q -deformation:

Axiom 3.1 (SED Cyclic Algebra).

$$\mathcal{A}_{\text{SED}} = \langle H_{st}, H_{ts} \mid H_{st}H_{ts} = q H_{ts}H_{st}, \quad q = e^{2\pi i\theta}, \quad |q| = 1 \rangle \quad (12)$$

where $\theta \in \mathbb{R}/\mathbb{Z}$ is the **dual twist angle**.

Physical regimes of θ :

- $\theta = 0$ ($q = 1$): standard commutative algebra, $[H_{st}, H_{ts}] = 0$, no SED asymmetry effects.
- $\theta = 1/2$ ($q = -1$): maximal non-commutativity, anti-commutation relation $\{H_{st}, H_{ts}\} = 0$, duality coupling strength maximized.
- $\theta = p/q$ (rational): **topologically protected quantization platforms**, dual twist becomes a discrete topological invariant.

3.4 The q -Deformed Commutator and Duality Norm

The q -deformed commutator:

$$[H_{st}, H_{ts}]_q = H_{st}H_{ts} - q H_{ts}H_{st} \quad (13)$$

The core dynamical order parameter—the **duality commutator norm**:

$$\mathcal{C} = \|[H_{st} \otimes H_{ts}, H_{ts} \otimes H_{st}]_q\| \quad (14)$$

Properties of $\mathcal{C}$:

- Dimension: $[\mathcal{C}] = \text{Energy}^2$, quantifying the intrinsic spacetime-dual self-referential coupling strength.
- At quantum critical points, $\mathcal{C}$ serves as an order parameter exhibiting singular scaling behavior.
- In the classical limit ($\hbar \rightarrow 0$, or equivalently $\theta \rightarrow 0$), $\mathcal{C} \rightarrow 0$ and SED reduces to standard quantum mechanics.

4 Spectroscopic Consequences: Intrinsic Asymmetry ϵ and Strict Distinction from Fano Effects

4.1 SED-Corrected Spectral Line Shape

The standard single-resonance Lorentzian:

$$L(\nu) = \frac{I_0}{1 + \left(\frac{\nu - \nu_0}{\Gamma/2}\right)^2} \quad (15)$$

Introducing the SED odd-part asymmetry correction, the complete spectral intensity:

$$I_{\text{SED}}(\nu) = I_0 \cdot L(\nu) \cdot \left[1 + \epsilon \cdot \mathcal{A}\left(\frac{\nu - \nu_0}{\Gamma/2}\right)\right] \quad (16)$$

where $\mathcal{A}(x)$ is an odd function (x, x^3 , etc.), and ϵ is the SED-exclusive spectral asymmetry parameter.

4.2 Integral Definition and Scaling of ϵ

$$\epsilon = \frac{\int_{\nu_0}^{\infty} I_{\text{SED}}(\nu) d\nu - \int_{-\infty}^{\nu_0} I_{\text{SED}}(\nu) d\nu}{\int_{-\infty}^{\infty} I_{\text{SED}}(\nu) d\nu} \quad (17)$$

Scaling relation to duality strength and linewidth:

$$\epsilon \sim \frac{\mathcal{C}}{\Gamma^2} = \frac{\| [H_{st} \otimes H_{ts}, H_{ts} \otimes H_{st}]_q \|}{\Gamma^2} \quad (18)$$

Core physical characteristics of ϵ :

- Real number, intrinsic and inherent;
- Persists in purely discrete systems without continuous coupling;
- Near critical points satisfies temperature scaling $\epsilon(T) \sim |T - T_c|^{\nu_z}$.

4.3 Fundamental Distinction from Fano Interference

Discrimination criterion 1: Sever the system's coupling channel to the continuum. Fano asymmetry disappears completely, but SED asymmetry ϵ persists, observable as spin decoherence time-domain asymmetric envelopes.

Table 1: Fundamental distinction between Fano interference and SED asymmetry

Feature	Fano q Interference	SED ϵ Asymmetry
Mathematical origin	Quantum path interference between discrete and continuum states	q -deformed commutator norm of spacetime-dual algebra
Domain	Complex number, contains interference phase	Real number, contributed solely by anti-Hermitian odd part
Continuum dependence	Requires continuous state coupling; asymmetry vanishes without continuum	Independent of external continuum coupling; persists in pure discrete closed systems
Coupling shutdown	External coupling $V \rightarrow 0$, line shape restores symmetry	Unaffected by external coupling strength; determined solely by spacetime duality strength $\mathcal{C}$

4.4 Vacuum Suppression and Quantum Critical Point Coherent Enhancement

In ordinary room-temperature isolated atomic systems, duality effects are Planck-scale suppressed:

$$\epsilon \sim \left(\frac{\ell_P}{\lambda}\right)^2 \approx 10^{-52} \quad (19)$$

experimentally unresolvable.

Near quantum critical points, the correlation length ξ diverges, and the number of coherently coupled particles N_{coh} within the coherent volume increases dramatically:

$$\mathcal{C}_{\text{eff}} \sim N_{\text{coh}} \cdot \mathcal{C}_0 \quad (20)$$

In the critical region, ϵ is elevated to the 10^{-10} – 10^{-6} detectable range. Conventional high-pressure classical compression cannot achieve equivalent coherent enhancement.

5 Quantum Critical Points: Spontaneous Exceptional Points and Odd-Even Critical Exponent Splitting

5.1 Spontaneous Exceptional Points (EPs) vs. Standard Non-Hermitian Systems

Conventional non-Hermitian optics and open quantum systems require fine artificial tuning of external gain and loss parameters to achieve simultaneous eigenvalue and eigenvector degeneracy at an Exceptional Point.

SED prediction: EPs are natural companions of quantum phase transitions, requiring no external loss/gain engineering. At the critical temperature T_c , the order parameter $\mathcal{C}(T) \rightarrow 0$, and the system spontaneously exhibits an EP where eigenstates coalesce and phase rigidity $r = |\langle \psi_L | \psi_R \rangle|$ vanishes.

Post-critical recovery follows:

$$r(T) \sim |T - T_c|^{\beta_{\text{odd}}} \quad (21)$$

and relaxation dynamics exhibits square-root singular scaling:

$$\delta A(t) \sim \sqrt{t} \quad (22)$$

Discrimination criterion 2: SED critical-point spontaneous EPs have no external parameter dependence; standard non-Hermitian EPs require fine field balancing. The two belong to distinct universality classes.

5.2 Odd-Even Critical Exponent Splitting: Breaking Fluctuation-Dissipation Constraints

Traditional critical theory and Fano frameworks are strictly constrained by the fluctuation-dissipation theorem, enforcing fixed relations between thermodynamic and optical response critical exponents:

$$\mu = \alpha + \nu(z - 2 + \eta) \quad (23)$$

SED introduces the anti-Hermitian odd part H_{odd} with anomalous dimension Δ_{odd} , correcting the optical conductivity imaginary-part critical exponent:

$$\mu_{\text{SED}} = \mu_{\text{standard}} + \nu \cdot \Delta_{\text{odd}} \quad (24)$$

Directly yielding measurable exponent splitting:

$$\boxed{\Delta\mu = \mu_{\text{odd}} - \mu_{\text{even}} \neq 0} \quad (25)$$

Discrimination criterion 3: Simultaneously measure specific heat and optical conductivity at a quantum critical point. If odd and even critical exponents exhibit non-negligible discrepancy, SED is confirmed—unexplainable by Fano or traditional critical phenomena.

6 Three Falsifiable Experimental Discrimination Protocols

6.1 Protocol I: Time-Domain Asymmetric Decoherence Envelope

System: Heavy-fermion compounds ($\text{CeCu}_{6-x}\text{Au}_x$, YbRh_2Si_2) near quantum critical point, nuclear spins or local electron spins.

Protocol: Ramsey interference sequence

$$\pi/2 \text{ pulse} \longrightarrow \text{free evolution } \tau \longrightarrow \text{second } \pi/2 \text{ pulse} \quad (26)$$

Measure spin-echo polarization:

$$P(\tau) = \frac{1}{2} (1 + \text{Re}[\rho_{12}(\tau)]) \quad (27)$$

SED signature:

$$\rho_{12}(\tau) = \rho_0 e^{-\Gamma_{\text{eff}}\tau/2} \cdot [1 + \epsilon \cdot \text{sgn}(\tau - \tau_0)] \quad (28)$$

The decoherence envelope exhibits different decay rates for $\tau < \tau_0$ and $\tau > \tau_0$; the characteristic time τ_0 diverges at T_c .

Exclusivity: Fano effects exist only in frequency-domain interference; cannot explain time-domain asymmetric spin relaxation.

6.2 Protocol II: Quantum Critical Point Odd-Even Critical Exponent Splitting

System: YBCO pseudogap regime, iron-based high- T_c superconductors, heavy-fermion antiferromagnetic quantum critical points.

Protocol: Synchronized thermodynamic and optical characterization

1. Low-temperature calorimetry: measure $C(T) \sim |T - T_c|^{-\alpha}$ (even Hermitian channel);
2. Far-infrared synchrotron pump-probe: extract $\text{Im} \sigma(\omega, T) \sim |T - T_c|^{-\mu}$ (odd non-Hermitian dual channel).

SED signature: Direct fitting yields α and μ ; verify that $\mu - \alpha$ deviates from the standard fluctuation-dissipation scaling relation, exhibiting stable non-zero $\Delta\mu$.

Exclusivity: Fano interference and traditional critical fluctuations strictly bind odd-even channel exponents; exponent splitting is forbidden.

6.3 Protocol III: Quantum Critical Point Spontaneous EP and $\sqrt{\tau}$ Relaxation Scaling

System: Optical microcavity exciton-polaritons, superconducting qubit arrays at Mott insulator-metal quantum phase transition.

Protocol: Phase rigidity and pump-probe relaxation dynamics

1. Phase rigidity: $r = \frac{|\langle \psi_L | \psi_R \rangle|}{\|\psi_L\| \|\psi_R\|}$;
2. Ultrafast pump-probe spectroscopy, recording transient absorption relaxation curve $\delta A(t)$.

SED signatures:

1. At $T = T_c$, phase rigidity r spontaneously approaches zero without external gain/loss fine-tuning;
2. Post-critical recovery satisfies power law $r(T) \sim |T - T_c|^{\beta_{\text{odd}}}$;
3. Transient relaxation dynamics exhibits square-root singular scaling $\delta A(t) \sim \sqrt{t}$.

Exclusivity: Standard non-Hermitian optical EPs require fine tuning of external loss/gain; cannot spontaneously appear at phase transitions with square-root relaxation.

7 Discussion: SED's Relation to Existing Theories

7.1 Compatibility and Innovation

1. **Loop Quantum Gravity:** SED's cyclic q -duality algebra is compatible with spin network braiding topological structures; the duality norm $\mathcal{C}$ may correspond to area quantum spectrum topological quantum numbers.
2. **String Theory / AdS-CFT Holographic Duality:** The $st \leftrightarrow ts$ spacetime dual map parallels holographic boundary-bulk duality; the twist angle θ may correspond to boundary conformal field theory topological twist parameters.

3. **Noncommutative Geometry (Connes):** SED’s q -deformed commutator is a fundamental realization of noncommutative geometry in spacetime degrees of freedom.
4. **Non-Hermitian Quantum Mechanics:** H_{odd} provides an intrinsic non-Hermitian microscopic origin independent of external open-system dissipation, remedying the inability of open quantum system theory to explain non-Hermitian effects in closed systems.
5. **Quantum Thermodynamics:** The dual cyclic tensor structure provides a microscopic mechanism for irreversible entropy increase; duality strength $\mathcal{C}$ serves as a novel quantum resource whose dissipation directly generates the thermodynamic arrow of time.

7.2 Open Problems

1. **Microscopic quantitative calculation:** For specific atomic and condensed matter systems, how to solve the duality commutator norm $\mathcal{C}$ from first principles?
2. **Twist angle topological correspondence:** The strict mathematical relation between dual twist angle θ and topological invariants / topological order.
3. **Strong gravity limit:** How does SED’s dual structure modify general relativistic classical gravity predictions in curved spacetime and strong gravitational fields, constructing SED’s complete low-energy limit of quantum gravity?

7.3 Conclusion

SED is not a supplementary correction to existing quantum optics or critical theory, but a thorough reconstruction of the ontological relation between spacetime and quantum dynamics. By replacing SEG’s spacetime-separation tensor $\Psi_s \otimes \Psi_t$ with the dual tensor $\Psi_{\text{SED}} = \Psi_{st} \otimes \Psi_{ts}$, the presumption of “pre-separated classical spacetime” is transformed into an emergent product of duality-relation dynamics.

SED’s predictions—intrinsic spectral asymmetry parameter ϵ , quantum critical point spontaneous Exceptional Points, and odd-even critical exponent splitting—all possess strictly calculable, discriminable, and falsifiable experimental characteristics, directly testable by existing condensed matter, ultrafast optics, and superconducting quantum experiments. If any of the three experimental signatures is observed and confirmed, it will fundamentally rewrite the underlying paradigms of quantum gravity, non-Hermitian physics, and quantum critical phenomena, marking the transition of theoretical framework from geometric dynamics to duality dynamics.

A FAQ: Common Referee Challenges and Responses

Q1: Is the SED asymmetry parameter ϵ merely an equivalent rewriting of the Fano q parameter?

A: The two are fundamentally inequivalent. Fano q is determined by discrete-continuum external coupling matrix elements; in pure closed discrete systems (trapped ions, cold atoms without radiation channels) coupling vanishes and the line shape is perfectly symmetric. SED ϵ is intrinsically determined by the spacetime-dual q -deformed commutator

norm $\mathcal{C}$, independent of external continuum channels. The closed-system time-domain decoherence asymmetry is a completely Fano-irreproducible observational signature.

Q2: Exceptional Points are already known in non-Hermitian physics; why does SED require a new framework?

A: Standard open non-Hermitian EPs depend on artificial fine-tuning of external gain and loss strengths, belonging to externally-driven effects. SED predicts EPs as intrinsic critical phenomena spontaneously driven by the quantum phase transition order parameter $\mathcal{C}(T)$, requiring no external loss/gain tuning. The two differ in physical origin and experimental conditions, belonging to distinct universality classes.

Q3: Is the temperature scaling of $\mathcal{C}$ merely a standard renormalization group (RG) consequence?

A: Traditional RG flows act only on the pure Hermitian Hamiltonian even part. SED introduces an independent RG flow for the dual-space anti-Hermitian odd part, generating an anomalous dimension Δ_{odd} that directly causes critical exponent splitting. This phenomenon violates traditional fluctuation-dissipation theorem constraints and does not belong to known RG framework standard results; it is uniquely new critical behavior.

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